

Multiple Choice (10 marks)

1. Which is a valid expression in Python 3.5?
 - (a) `sort('ab')`
 - (b) `sorted('ab')`
 - (c) `"ab".sort()`
 - (d) `1/0`
2. Which of the following best explains what happens when a new device is connected to the Internet?
 - (a) A device driver is assigned to the device.
 - (b) An Internet Protocol (IP) address is assigned to the device.
 - (c) A packet number is assigned to the device.
 - (d) A Web site is assigned to the device.
3. Let `l = [1,2,3,4]`. What is `sum(l)` in Python 3?
 - (a) 4
 - (b) 1
 - (c) 10
 - (d) 8
4. In Python 3, what is `['a', 'Chemistry', 0, 1][-3]`?
 - (a) a
 - (b) Chemistry
 - (c) 0
 - (d) 1
5. Which of the following spreadsheet functions would be most useful for detecting improbably high or low values that have become part of a data set as a result of data entry errors?
 - (a) A function that averages numeric values in a column or row
 - (b) A function that counts the values in a column or row
 - (c) A function that rounds a numeric value
 - (d) A function that sorts values in a column or row

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, using `b[::2]` on the list `[11,13,15,17,19,21]` returns `[19,21]`.
7. True or False: $O(\log n)$ grows faster than $O(n^2)$ as the input size increases.
8. True or False: In Python 3, the `random()` function sets the integer starting value used in generating random numbers.
9. True or False: In Python 3, shifting the number 1 to the left by 3 bits results in the value 8.
10. True or False: In Python 3, the `strip()` function removes all leading and trailing whitespace from a string.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the `'isupper()'` function in Python 3 determines if all characters in a string are uppercase. Provide examples to illustrate its usage and potential applications.
12. Explain how the lambda function `r = lambda q: q * 2` works in Python 3, and analyze the output of `r(3)`. What does this reveal about the use of lambda functions?

End of Paper